

Microeconomics II – Class 16

Consumer's intertemporal choice



Class summary

- * Intertemporal choice
- * Budget constraint
 - * with no interest earned on savings and borrowing not allowed
 - * with interest earned on savings and borrowing allowed
- * Present and future values
- * Comparative statics: interest rate changes
- * Slutsky equation and intertemporal choice

- * Readings: Varian (cap. 10)



Intertemporal choice: Budget constraint

- * Intertemporal choice:
 - * decision about the amount of consumption in each period of time
- * Budget constraint

$C \equiv (c_1, c_2)$: quantity consumed in periods 1 and 2.

$P \equiv (p_1, p_2)$ prices in periods 1 and 2.

We will assume that $(p_1, p_2) = (1, 1)$ (no inflation)

$M \equiv (m_1, m_2)$: endowment of income in periods 1 and 2.



Budget constraint with no interest earned on savings and borrowing not allowed

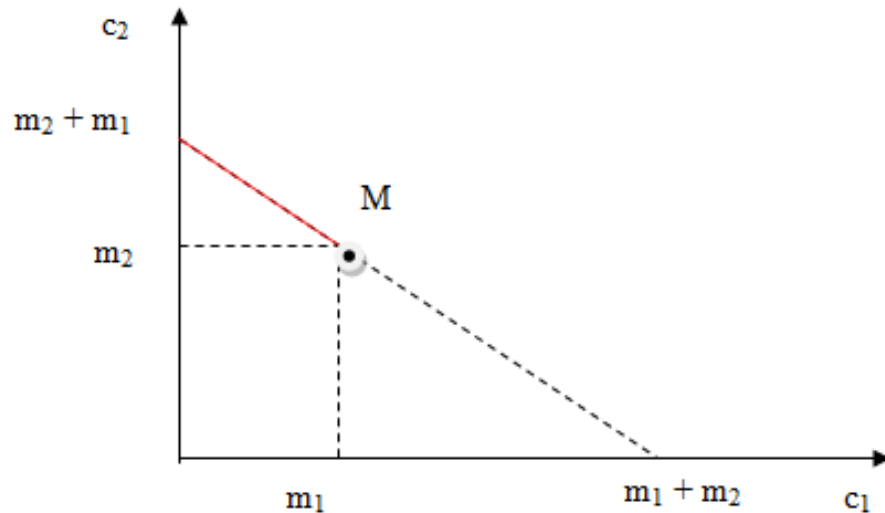
- * Period 1's choice:
 - * Consume the entire income ($c_1 = m_1$) or save a part for the following period ($c_1 < m_1$)
- * Quantity of consumption in period 2: $c_2 = m_2 + (m_1 - c_1)$
- * Slope of the budget constraint: $\frac{\partial c_2}{\partial c_1} = -1$



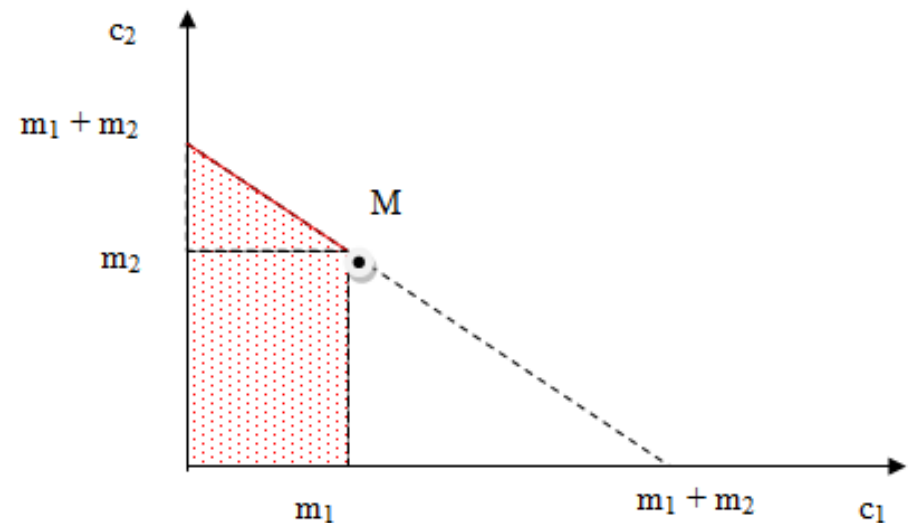
Budget constraint with no interest earned on savings and borrowing not allowed

* Graphical representation:

Budget constraint



Budget set



Budget constraint with interest earned on savings and borrowing allowed

$c_1 < m_1 \Rightarrow$ Lender

Making a deposit in a bank is an example of lending money

$c_1 > m_1 \Rightarrow$ Borrower

Someone who takes out a loan from a commercial bank, for example

r : interest rate

We will assume the same interest rate is applied to loans and deposits



Budget constraint with interest earned on savings and borrowing allowed

* **Lender** $(c_1 < m_1)$

$$c_2 = m_2 + (m_1 - c_1) + r \times (m_1 - c_1) \Leftrightarrow$$

Deposits earn interest

$$\Leftrightarrow c_2 = m_2 + (1+r) \times (m_1 - c_1)$$

* **Borrower** $(c_1 > m_1)$

$$c_2 = m_2 - (c_1 - m_1) - r \times (c_1 - m_1) \Leftrightarrow$$

Payment of interest and principal
(capital repayment)

$$\Leftrightarrow c_2 = m_2 + (1+r) \times (m_1 - c_1)$$

* **The result is the same! Budget constraint:**

$$c_2 = m_2 + (1+r) \times (m_1 - c_1)$$

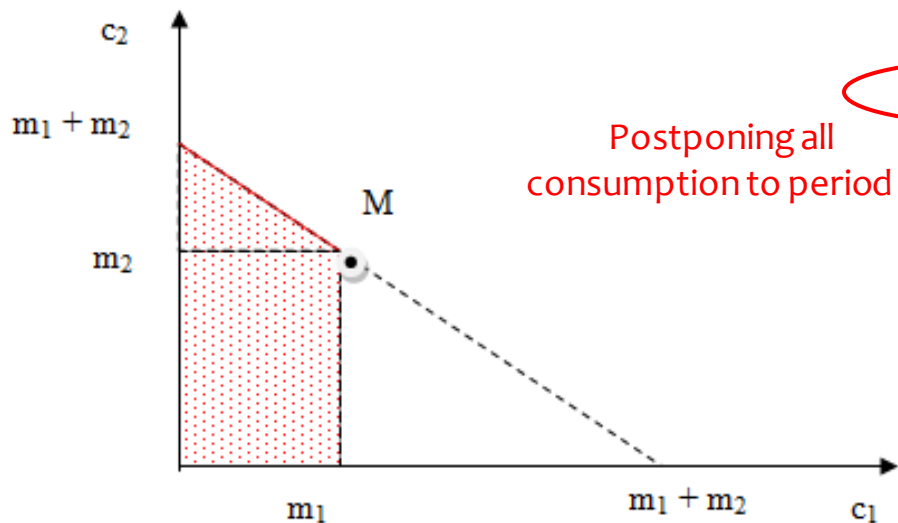
Slope: $\frac{\partial c_2}{\partial c_1} = -(1+r)$



Budget constraint with interest earned on savings and borrowing allowed

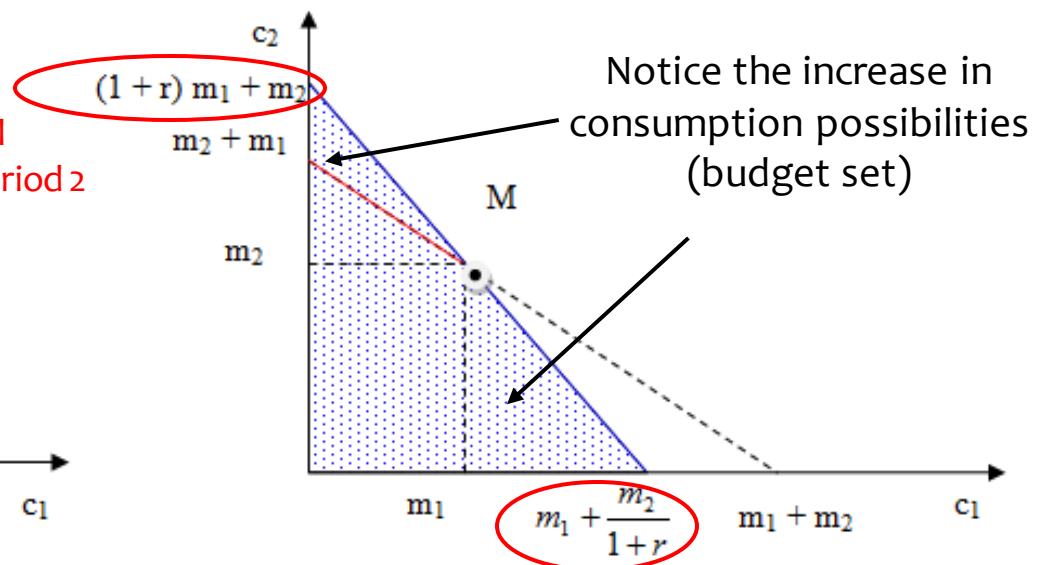
* Graphical representation:

Previous budget constraint and budget set



Postponing all consumption to period 2

New budget constraint and budget set



Advancing all consumption to period 1



Present and Future Value

- * If $c_1 = 0 \Rightarrow c_2 = m_2 + (1+r) \times m_1 \equiv$ future value of the income endowment, maximum feasible consumption level in period 2

- * If $c_2 = 0 \Rightarrow c_1 = m_1 + m_2 / (1+r) \equiv$ present value of the income endowment, maximum feasible consumption level in period 1

- * Future value budget constraint:

$$c_2 = m_2 + (1+r) \times (m_1 - c_1) \Leftrightarrow c_1 \times (1+r) + c_2 = m_1 \times (1+r) + m_2$$

- * Present value budget constraint

$$c_2 = m_2 + (1+r) \times (m_1 - c_1) \Leftrightarrow c_1 + \frac{c_2}{(1+r)} = m_1 + \frac{m_2}{(1+r)}$$



Present and Future Value

- * Future value budget constraint

$$c_1 \times (1+r) + c_2 = m_1 \times (1+r) + m_2$$

- * Present value budget constraint

$$c_1 + \frac{c_2}{(1+r)} = m_1 + \frac{m_2}{(1+r)}$$

- * Both equations have this form: $p_1 x_1 + p_2 x_2 = p_1 m_1 + p_2 m_2$

$$FV : \begin{cases} p_1 = 1+r \\ p_2 = 1 \end{cases} \leftarrow \begin{array}{l} \text{Future value of the} \\ \text{opportunity cost of} \\ \text{consuming in the present} \end{array}$$

$$PV : \begin{cases} p_1 = 1 \\ p_2 = \frac{1}{1+r} \end{cases} \leftarrow \begin{array}{l} \text{Present value of the} \\ \text{opportunity cost of} \\ \text{consuming in the future} \end{array}$$



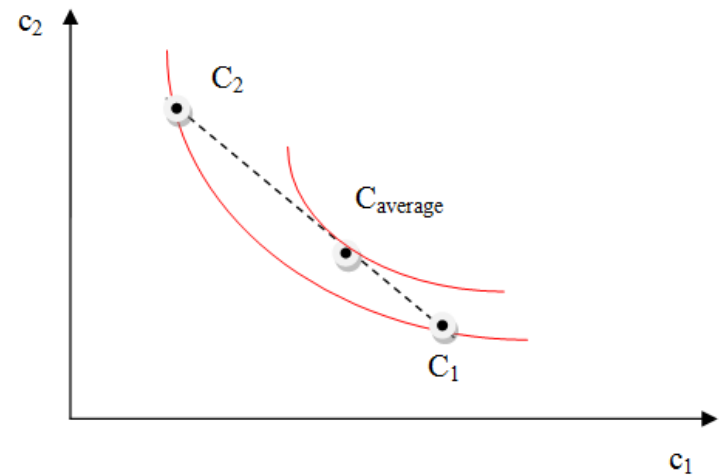
Intertemporal consumer choice

- * Consumer's problem

$$\begin{cases} \max_{c_1, c_2} U(c_1, c_2) \\ \text{s.t. } c_1 + \frac{c_2}{(1+r)} = m_1 + \frac{m_2}{(1+r)} \end{cases}$$

- * We will assume convex indifference curves

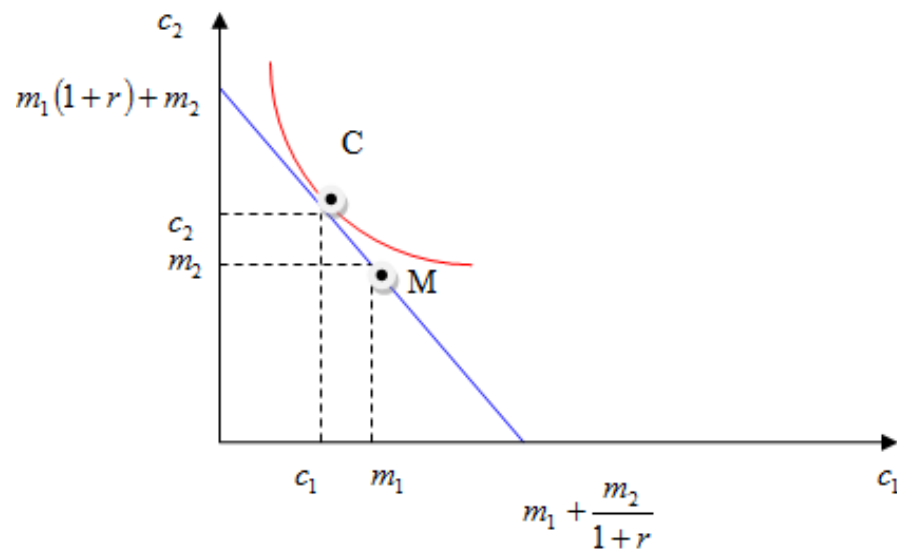
- * the consumer prefers to distribute consumption by both periods instead of concentrating it in a single period of time.



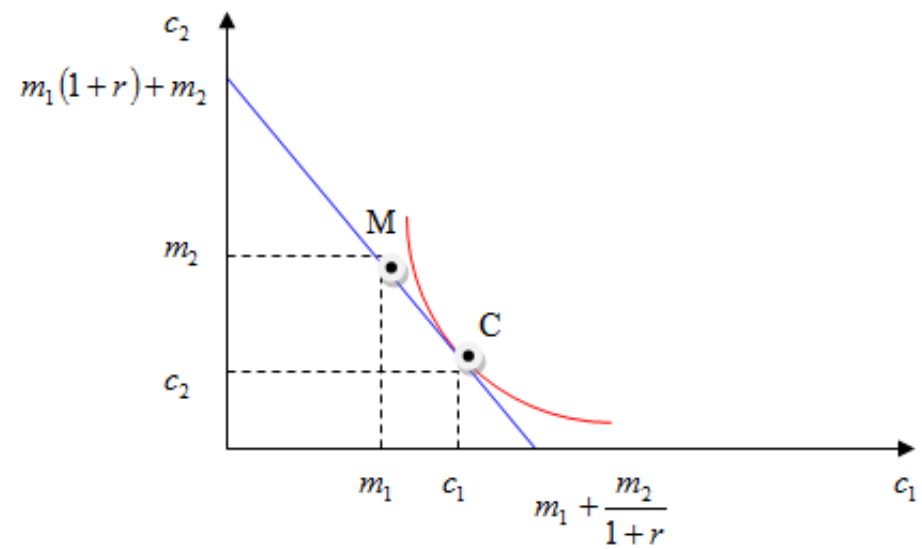
Intertemporal consumer choice

* Graphical representation:

Lender ($c_1 < m_1$)

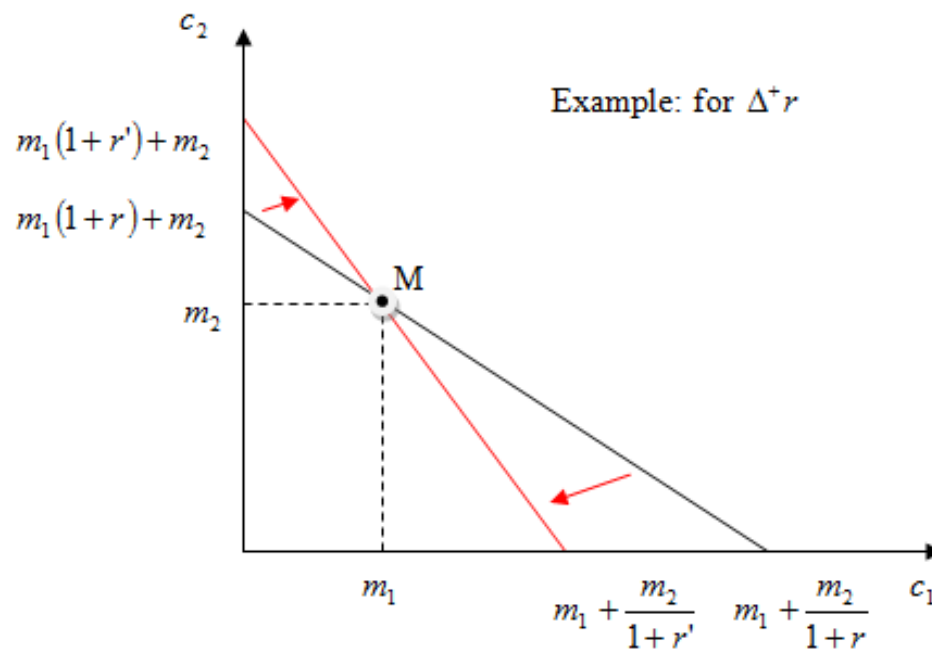


Borrower ($c_1 > m_1$)



Comparative statics: interest rate changes

- * Change in the interest rate from r to r'



M is still on the budget constraint.

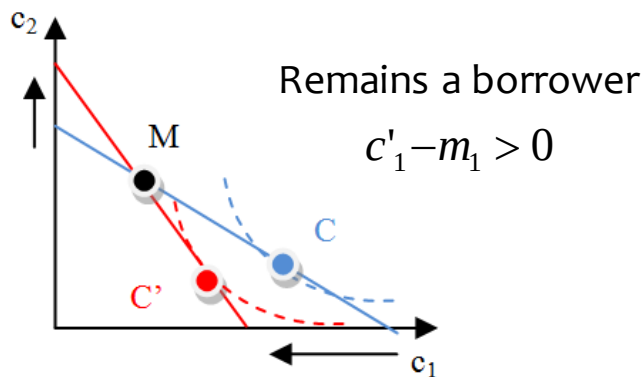
The future and present values of the income endowment are affected by the changes in interest rates.

Where the new optimal consumption bundle will be depends on whether the consumer was a lender or a borrower before the change in interest rate.

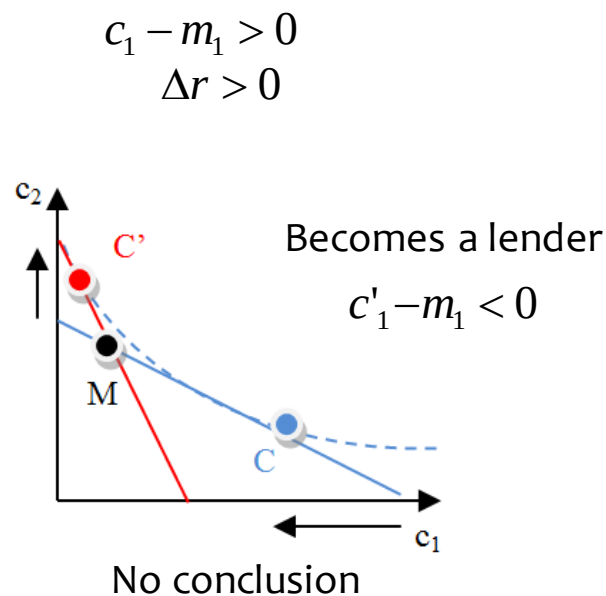


Comparative statics: interest rate changes

- * Is the consumer better or worse off with the interest rate change?
- * Let's check all possible cases
 - * Baseline situation: borrower
 - * Interest rate increases



He's worse off (proof by the principle of revealed preferences)

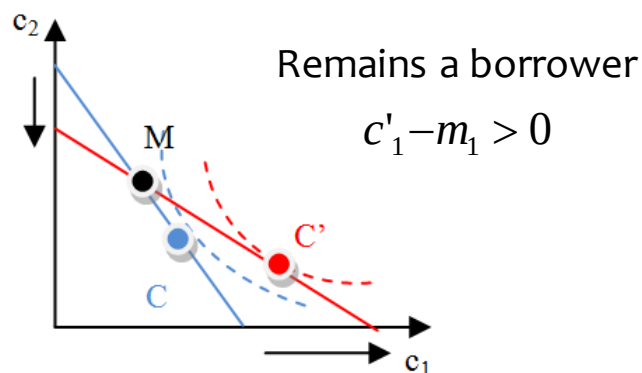


No conclusion

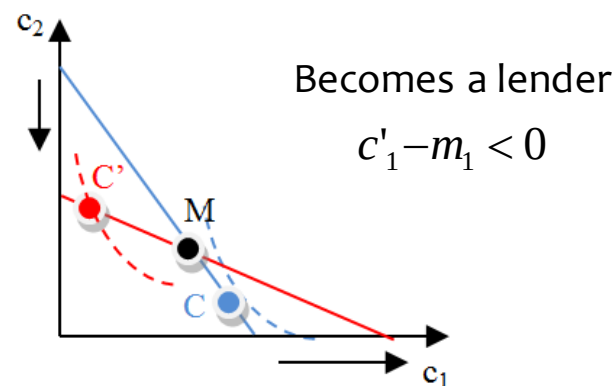


Comparative statics: interest rate changes

- * Is the consumer better or worse off with the interest rate change?
- * Let's check all possible cases
 - * Baseline situation: borrower
 - * Interest rate decreases



He's better off (proof by the principle of revealed preferences)



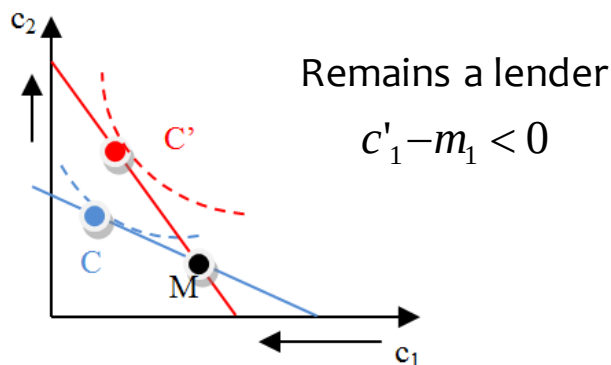
He's worse off (proof by the principle of revealed preferences)

Does not occur if the consumer is rational

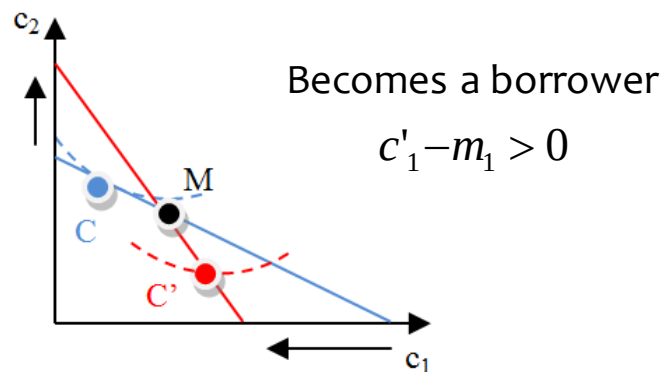


Comparative statics: interest rate changes

- * Is the consumer better or worse off with the interest rate change?
- * Let's check all possible cases
 - * Baseline situation: lender
 - * Interest rate increases



He's better off (proof by the principle of revealed preferences)



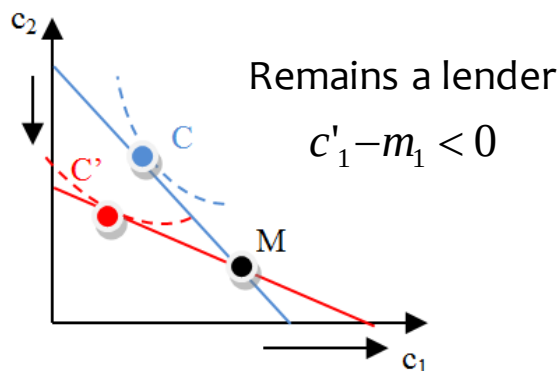
He's worse off (proof by the principle of revealed preferences)

Does not occur if the consumer is rational



Comparative statics: interest rate changes

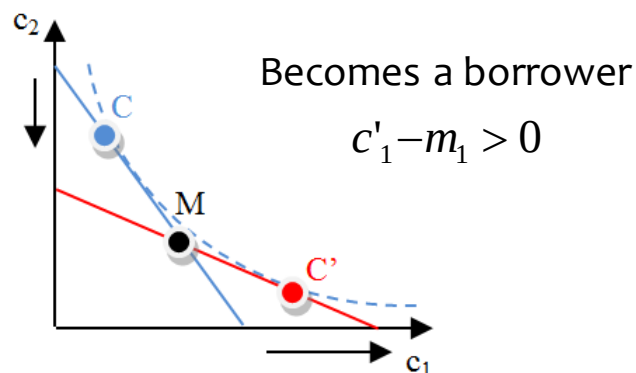
- * Is the consumer better or worse off with the interest rate change?
- * Let's check all possible cases
 - * Baseline situation: lender
 - * Interest rate decreases



He's worse off (proof by the principle of revealed preferences)

$$c_1 - m_1 < 0$$

$$\Delta r < 0$$



No conclusion



Comparative statics: interest rate changes

- * *What can we conclude from the previous analysis?*
- * *Conclusion A: a borrower does not become a lender due to a fall in the interest rate.*
- * *Conclusion B: a lender does not become a borrower due to a rise in the interest rate.*



Slutsky equation and intertemporal choice

- * What is the total effect of increasing the interest rate on the consumptions of periods 1 and 2?

* Future value budget constraint: $c_1 \times (1+r) + c_2 = m_1 \times (1+r) + m_2$

* Present value budget constraint: $c_1 + \frac{c_2}{(1+r)} = m_1 + \frac{m_2}{(1+r)}$

Future value of the opportunity cost of consuming in the present $p_1 = 1+r$

Present value of the opportunity cost of consuming in the future $p_2 = \frac{1}{1+r}$

$$\Delta^+ r \Leftrightarrow \begin{cases} \Delta^+ p_1 \\ \Delta^- p_2 \end{cases}$$

$$\Delta^- r \Leftrightarrow \begin{cases} \Delta^- p_1 \\ \Delta^+ p_2 \end{cases}$$



Slutsky equation and intertemporal choice

- * What is the total effect of increasing the interest rate on the consumptions of periods 1 and 2?

- * Slutsky equation for
intertemporal consumption:

$$\frac{dc_1}{dp_1} = \frac{\partial c_1^s}{\partial p_1} + (m_1 - c_1) \frac{\partial c_1^m}{\partial m}$$

$\frac{dc_1}{dp_1}$ Total effect (ambiguous sign in the presence of an endowment, even for normal goods)

$\frac{\partial c_1^s}{\partial p_1}$ Substitution effect (the sign is opposite to the sign of the interest rate change)

$-\frac{\partial c_1^m}{\partial m} c_1$ Ordinary income effect (sign opposite to the sign of the interest rate change if we assume consumption is normal regarding income changes)

$\frac{\partial c_1^m}{\partial m} m_1$ Endowment income effect (sign identical to the sign of the interest rate change if we assume consumption is normal regarding income changes)



Slutsky equation and intertemporal choice

- * What is the total effect of increasing the interest rate on the consumptions of periods 1 and 2?

$$\Delta^+ r \Leftrightarrow \begin{cases} \Delta^+ p_1 \\ \Delta^- p_2 \end{cases} \quad \Delta^- r \Leftrightarrow \begin{cases} \Delta^- p_1 \\ \Delta^+ p_2 \end{cases}$$

- * We assume consumption is normal regarding income changes: $\frac{\partial c_1^m}{\partial m} > 0$

- * Slutsky equations:
$$\begin{cases} \frac{dc_1}{dp_1} = \frac{\partial c_1^s}{\partial p_1} + (m_1 - c_1) \frac{\partial c_1^m}{\partial m} \\ \frac{dc_2}{dp_2} = \frac{\partial c_2^s}{\partial p_2} + (m_2 - c_2) \frac{\partial c_2^m}{\partial m} \end{cases}$$

- * The total effect depends on whether the consumer was initially a borrower or a lender

$$m_1 - c_1 > 0?$$

$$m_1 - c_1 < 0?$$



Slutsky equation and intertemporal choice

- * What is the total effect of increasing the interest rate on the consumptions of periods 1 and 2?

$$\Delta^+ r \Leftrightarrow \begin{cases} \Delta^+ p_1 \\ \Delta^- p_2 \end{cases}$$

$$\text{Borrower: } \begin{cases} m_1 - c_1 < 0 \\ m_2 - c_2 > 0 \end{cases}$$

$$\text{Lender: } \begin{cases} m_1 - c_1 > 0 \\ m_2 - c_2 < 0 \end{cases}$$

$$\begin{cases} \frac{dc_1}{dp_1} = \frac{\partial c_1^s}{\partial p_1} + (m_1 - c_1) \frac{\partial c_1^m}{\partial m} \\ \frac{dc_2}{dp_2} = \frac{\partial c_2^s}{\partial p_2} + (m_2 - c_2) \frac{\partial c_2^m}{\partial m} \end{cases}$$

$$\begin{cases} \frac{dc_1}{dp_1} = \frac{\partial c_1^s}{\partial p_1} + (m_1 - c_1) \frac{\partial c_1^m}{\partial m} \\ \frac{dc_2}{dp_2} = \frac{\partial c_2^s}{\partial p_2} + (m_2 - c_2) \frac{\partial c_2^m}{\partial m} \end{cases}$$

$$\Delta^- r \Leftrightarrow \begin{cases} \Delta^- p_1 \\ \Delta^+ p_2 \end{cases}$$

$$\text{Borrower: } \begin{cases} m_1 - c_1 < 0 \\ m_2 - c_2 > 0 \end{cases}$$

$$\text{Lender: } \begin{cases} m_1 - c_1 > 0 \\ m_2 - c_2 < 0 \end{cases}$$

$$\begin{cases} \frac{dc_1}{dp_1} = \frac{\partial c_1^s}{\partial p_1} + (m_1 - c_1) \frac{\partial c_1^m}{\partial m} \\ \frac{dc_2}{dp_2} = \frac{\partial c_2^s}{\partial p_2} + (m_2 - c_2) \frac{\partial c_2^m}{\partial m} \end{cases}$$

$$\begin{cases} \frac{dc_1}{dp_1} = \frac{\partial c_1^s}{\partial p_1} + (m_1 - c_1) \frac{\partial c_1^m}{\partial m} \\ \frac{dc_2}{dp_2} = \frac{\partial c_2^s}{\partial p_2} + (m_2 - c_2) \frac{\partial c_2^m}{\partial m} \end{cases}$$



Slutsky equation and intertemporal choice

* Summary-table:
$$\frac{dc_1}{dp_1} = \frac{\partial c_1^s}{\partial p_1} + (m_1 - c_1) \frac{\partial c_1^m}{\partial m}$$

Effects of a change in the interest rate	$\Delta^+ r$	$\Delta^- r$
<i>Lender</i> $m_1 - c_1 > 0$ $m_2 - c_2 < 0$	indeterminate	$\Delta^+ c_1 \wedge \Delta^- c_2$
<i>Borrower</i> $m_1 - c_1 < 0$ $m_2 - c_2 > 0$	$\Delta^- c_1 \wedge \Delta^+ c_2$	indeterminate



Slutsky equation and intertemporal choice

- * Conclusions:
- * Conclusion A:
 - * *a **borrower** will decrease consumption in period 1 (which was being financed by the loan) and increase it in period 2 when faced with a **rise in the interest rate**.*
 - * *due to the increase in the future value of present consumption's opportunity cost the consumer will consume less in the present and more in the future.*
 - * *The future benefit lost with every unit borrowed in the present is higher.*
- * Conclusion B:
 - * *a **lender** will increase consumption in period 1 and decrease it in period 2 (which was being financed by prior savings) when faced with a **decrease in the interest rate**.*
 - * *due to the decrease in the future value of current consumption's opportunity cost, the consumer will consume more in the present and less in the future.*
 - * *The future benefit obtained with every unit saved in the present is lower.*

